MID-II Engineering Physics Short Answer Questions

S.No	UNIT-III	CO	Marks
	Chapter-1 Dielectrics	CO3	2M
1	Explain polar and non-polar dielectric	CO3	2M
2	Define electric polarization and Dielectric displacement	CO3	2M
3.	Derive D,E,P relation	CO3	2M
4.	Show that $\chi_e=\epsilon_r-1$	CO3	2M
5.	Mention the application of dielectric materials	CO3	2M
6.	Derive Claussius- Mosotti Equation	CO3	2M
	Chapter-2 Magnetic Materials		
.7.	Define the terms a)Magnetic Permeability b) Magnetic Susceptibility C) Magnetization	CO3	2M
8.	Obtain the relation between magnetic flux density, magnetization and magnetizing field	CO3	2M
9.	Difference between soft and hard magnetic materials	CO3	2M
10.	Show that $\mu=1+\chi$	CO3	2M
11.	What is Bohr Magneton?	CO3	2M
	UNIT-IV Chapter-1 Quantum Mechanics		
1	Write some properties of matter waves	CO4	2M
2	State and explain Heisenberg principle	CO4	2M
3	Write the properties of Wave function	CO4	2M
4.	Expalin de Broglies hypothesis	CO4	2M
	Chapter-2 Free Electron Theory		
5	What is Fermi-energy level? Give mathematical expression for it	CO4	2M
6	What is the significance of Fermi-Dirac distribution function?	CO4	2M
7	Assumptions of quantum free electron theory	CO4	2M
	UNIT-V Semiconductors		
1	Define Drift and Diffusion currents in semiconductors.	CO5	2M
2	State Hall effect with diagram.	CO5	2M
3	Classify the different types of semiconductors.	CO5	2M
4	Define Fermi energy level. Give mathematical expression	CO5	2M
5	Write the applications of Hall-effect.	CO5	2M
6	Compare Intrinsic and Extrinsic semiconductors	CO5	2M
7	Compare p type and n type semiconductors	CO5	2M